Welcome to FORS2_impol_redux

Purely python based back-to-back reducer of FORS2 impol data

In this README file I give a short description on what the pipeline does, how to install it and how to make it run.

Introduction:

The pipeline is designed to do on-the-fly reduction of FORS2 imaging polarimetry of a point source. While the results of the pipeline are not directly publishable, they should provide a good, quick estimate on how strongly polarised the source appears to be.

The pipeline is designed to take in FORS2 impol data of one target for one or more nights. It searches for the needed files (SCI and BIAS) in the RAW_DIR and sorts them out per night. *No input parameters are needed!*

The pipeline also gives a rough estimate of the galactic interstellar polarisation (ISP) based on "stars" present in the field of view. While the estimated ISP is less reliable than the results for the specified target, the spread of found stars on the produced Q-U plane will help determine if the target is inherently polarised or if the polarisation signal is produced by dust grains in the Milky Way. However, note that the stars are identified using very simple FWHM and SNR cuts and not all of them may actually be stars or some might be saturated etc. As such the results of individual stars should be taken with a pinch (or a pint) of salt and if less than ~10 stars were found altogether the ISP is unreliable.

IMPORTANT

The pipeline performs instrumental correction following González-Gaitán+2020. It tries to perform the instrumental correction using the actual maps first, but if they are not downloaded, it resorts to using the presented simplifying equations. If you want to download the maps, you can do so here by choosing the files ending in inlaQ.fits and inlaU.fits for each of the filters you are using (BVRI supported). Then you need to edit inla_dirc variable on row 33 of the code to to point to the download directory.

"Installation"

1) Save the FORS2_impol_redux python script in directory of your choosing

2) Create a conda environment to ensure the correct package version are installed
   -- use the provided FORS2_impol_redux.yml environment file
   -- conda env create -f FORS2_impol_redux.yml

3) Activate the environment
   -- conda activate FORS2_impol

4) Open FORS2_impol_redux
   -- Change #!/Users/miipu/opt/anaconda3/bin/python at the top to your python dir
   -- Can be found with command which python in terminal
   -- Do this from within the conda environment!

5) run chmod +x FORS2_impol_redux to make it executable

6) Add the directory where you stored the script to your PATH (if not already added)
-- e.g. add export PATH="/path/to/dir/:$PATH to your .bashrc (or .bash_profile, .zprofile or .zshrc)

7) source .bashrc or open a new terminal window to load the changes

Known issues with installation:

-- Sometimes step 2) seems to add hidden characters to the python path and the directory is not found. If this happens try to retype the python path yourself and see if it fixes it.

-- If you get following error see Issue #2 for the fix
TypeError: zoom() got an unexpected keyword argument 'grid_mode'

-- If you need to remove the conda environment for any reason you can do it with
conda remove --name FORS2_impol --all

Overview of the pipeline:

First Half: Reduction

1) Searches for raw files in RAW_DIR (default: archive/)
-- SCI and corresponding BIAS files are needed
-- If target name (optional parameter) was not provided, the pipeline will prompt you to choose the correct object from list
-- The pipeline prints out the found science files for each setup with the angle and exp time. Check that the exposure times are the same for each angle under the setup.

2) Sort the files by night and find corresponding SCI and BIAS frames
-- If exposure times for all files for a night-filter-chip setup are not the same, user prompt is required
-- Identify the files that have different exposure time and they are moved to RAW_DIR/bad_fits/
-- Make sure you identify the correct ones as if bad ones are carried though the script will either crash or the results won't make sense!

3) Creates directories under OUT_DIR (default: FORS2_impol/*)

4) Creates Master Biases based on the bias files
--Saved in <OUTDIR>/<night>/bias/

5) Subtracts the Master biases from the SCI files (both CHIP1 and CHIP2) and performs cosmic ray removal with astroscrappy
-- SCI files saved in <OUTDIR>/<night>/sci/

6) Joins the two FORS2 CHIPS into *comb.fits files that are then used for the analysis
-- Saved in <OUTDIR>/<night>/sci/

-- Also deletes the CHIP specific files from step 5)

Second Half: Analysis

7) Finds sources in the image with photutils and estimates FWHM based on them
-- Finds the wanted source in specified e-beam pixel (default pixel = 834,780) and estimates FWHM for it
-- Finds the sources with similar or smaller FWHM. The used FWHM is then determined as the average of these objects, excluding the target unless it is the only found source in the image (unlikely)
-- These found "stars" are the basis for the ISP after a S/N cut (default >200).

8) Runs aperture photometry using photutils and saves results to OUT_DIR/<night>/cats/
-- Default settings work mostly fine, but several tweak options here. See Section *Optional parameters* for details

9) Solves for the QUPX and saves results (pdf and asci) in OUT_DIR/<night>/results/
-- Performs Chromatic Correction and Instrumental Correction if band is one of the following: u_HIGH, b_HIGH, v_HIGH, R_SPECIAL, I_BESS
-- If not in the list the stars cannot be trusted!

Running the pipeline:

1) First download the files from ESO archive: http://archive.eso.org/eso/eso_archive_main.html. I do this normally in the following way:
-- Use coordinates of the target
-- Polarimetry: FORS2/VLT
-- Category: Science
-- Specify a night (if preferred)
-- Download all the files. I download them as zip file
-- *Make sure you download only impol observations (impol) and not specpol (pmos)*
-- As biases are needed you can either "Run Association" on the raw calibration files and load all of them. This downloads a lot of unneeded files, but is the quickest way to get what you need. Alternatively, you can find the raw BIAS files yourself. This is slower but you won't waste space.

2) Unzip the downloaded files and run uncompress archive/*.Z

3) Run the pipeline

Recommended input:

FORS2_impol_redux -t <TARGET>

-- <TARGET> is the name of target in provided in the fits files ("OBJECT") and shown in ESO archive
-- if <TARGET> is not provided, pipeline will prompt you to choose the target from a list

-- The default setup works perfectly well in most cases, but sometimes specifications are needed (see below)
-- Note that the pipeline looks for the raw files by default in archive/

4) After the first run the pipeline no longer requires the raw SCI and BIAS files (unless something is wrong with the reduction), so they can be removed

Optional parameters:

With the optional parameters you can tweak the reduction procedure

--target_name	Target name found in fits files under "OBJECT"
--night <NIGHT>	Choose a specific night
--band	Chose Band (e.g. B, V, R, I)
--raw_dir	RAW_DIR directory (default archive/)
--out_dir	OUT_DIR directory (default FORS2_impol/)
--do_sci	Force to reduce the SCI files again
--do_bias	Force to reduce the BIAS files again
--do_phot	Run the aperture photometry again
--do_redux	Forces the complete reduction from raw files
--apers <aper1> <aper2> …	Choose the aperture sizes apertures to run (the values are multiplied by FWHM)
--xye	e-beam (lower of the two identical stripes) pixel location of the wanted source (default: 833,778)
--fwhm <FWHM>	Force a FWHM for a set of observations
--star_snr	SNR cut for the field stars. Default 200
--star_fwhm	FWHM cut for the field stars. Given as i: FWHM_star < i x FWHM_target. Default 1.3
--check_files	Investigate the found RAW files manually. Gives a prompt to approve the set of files for night-filter-chip based on the exposure times. Options: "all", "some", "none". All asks you to verify every setup, some only when the exposure times are not the same or number of files is wrong (not multiple of 4) and "none" assumes the files are good. Default "some".
--plot_stamps	Plot stamps of the apertures for each cycle
--bkg_type	Background type: 2dmap or annulus. 2dmap (default) appears to be better in general (especially if source is in extended galaxy), but annulus works also well in quiet environment
--r_annu	If using bkg_type=annulus, set the inner and outer radii of the said annulus. As default use the size of the largest aperture request + 5px
--save_bkg	Option to save the created 2d bkg map. It's used for source detection but it's also subtracted from the science image if bkg_type=2dmap.

Example:

FORS2_impol_redux -t SLSN --do_bias --do_sci --do_phot --night 2017-09-22 --apers 1.0 1.5 2.0 --fwhm 4.0 --xye 860 940 --bkg_type annulus

Forces the full reduction on of a target found at xy=860,940 on night 2017-09-22 with FWHM=4 for aperture sizes 1, 1.5 and 2 x FWHM using a background annulus

OUTPUT:

As a final product following files are found in OUT_DIR/<night>/results/:

log_target_<FILTER>.txt: asci files of the identified target for each filter. If the results you get look crazy, have a look at these files to see if the same x y pixels have been found for every o and e beam image (over the cycles and the four angles). If this is not the case, a wrong source was found for and results should not make sense.

QUPX_<FILTER>_target_<APER>.txt: QUPX + xy coordinates and SNR per each filter and chosen aperture for the target

QUPX_<FILTER>_stars_<CYCLE>.txt: QUPX + xy coordinates and SNR per filter and cycle for field stars found in the images

QU_plane_<FILTER>_<APER>.pdf: QU_plane of the results. Target and the field stars as well the galactic ISP estimate based on the field stars are shown. The ISP estimate is rough, but if +10 stars are found and the target is clearly offset from the ISP (and the stars) on the plane, the polarisation of the target is likely not caused by Galactic dust and it is probably either due to host galaxy dust or inherent polarisation of the target in question.

Known issues

-- Sometimes the correct target is not been identified for every *comb.fits file. This leads to unrealistic (e.g. high Q/U). Easy way to identify if this is an issue is to check the log_target*.file and QUPX_<FILTER>_target_<APER>.txt. The issue occurs most likely because sometimes the target is not positioned in the centre of the CCD. In this case you'll have to open the science files and identify the pixel coordinates of the target for the e-beam (the lower of the two identical stripes) and feed it as input: FORS2_impol_redux <TARGET> -xye <x-coord> <y-coord>

-- If different nights have a different target name in the fits file, you'll have to run the pipeline with both names separately

-- In rare occasions the location of the target can change slightly between consecutive cycles and this can lead to misidentification of the correct source. Have a look at the SCI files in case the locations between cycles seem to vary. There's no quick fix for this but if the offset is <5px you can try to choose px location halfway between with FORS2_impol_redux <TARGET> -xye <x-coord> <y-coord>

-- If the source is in a crowded field the pipeline takes a while to do the aperture photometry for all the sources. Just hang on with it.

-- If spectral polarimetry (PMOS) files are downloaded, the pipeline will crash.

-- In case the target is faint and/or in a bright galaxy, the FWHM estimation might not result in a good value. If this is the case, inspect the QUPX_<FILTER>_stars_<CYCLE>.txt file to see the FWHM of identified stars in the field and make an educated guess using --fwhm <FWHM>

Bugs and suggestions etc.

-- Bugs are likely to occur. If you encounter one, please let me know! Include the description of what you ran including the exact command and the error message.
-- In case you have suggestions on what you'd like me to add to the pipeline, let me know!
-- Preferably add them as issues in GitHub
-- You can also email me at miika.pursiainen@warwick.ac.uk

FORS2_impol_redux -h

```
Back-to-back reducer of FORS2 impol data

optional arguments:
  -h, --help            show this help message and exit
  -t TARGET, --target TARGET
                        The name of the science target in the fits files
  -n NIGHT, --night NIGHT
                        Specify wanted night (YYYY-MM-DD). Default "all".
  -b BAND, --band BAND  Specify wanted band. i.e. V_Bes. Default "all".
  -rd RAW_DIR, --raw_dir RAW_DIR
                        The directory of the raw files (default archive/)
  -od OUT_DIR, --out_dir OUT_DIR
                        The directory of the out files (default FORS2_impol/)
  -ds [DO_SCI [DO_SCI ...]], --do_sci [DO_SCI [DO_SCI ...]]
                        Forces reduction of science files in case already done
  -db [DO_BIAS [DO_BIAS ...]], --do_bias [DO_BIAS [DO_BIAS ...]]
                        Forces reduction of master bias in case already done
  -dp [DO_PHOT [DO_PHOT ...]], --do_phot [DO_PHOT [DO_PHOT ...]]
                        Forces the aperture photometry
  -dr [DO_REDUX [DO_REDUX ...]], --do_redux [DO_REDUX [DO_REDUX ...]]
                        Redo whole reduction precedure!
  -a APERS [APERS ...], --apers APERS [APERS ...]
                        The apertures (i) to use. Defined as i x FWHM
  -xye XYE [XYE ...], --xye XYE [XYE ...]
                        xy coordinates of e beam target. Default normally works, but the target is not
always in the middle!
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-fw FWHM, --fwhm FWHM
Forces fwhm (px). As default estimated based on sources available in the field
-ss STAR_SNR, --star_snr STAR_SNR
SNR cut for the field stars. Default 200
-sf STAR_FWHM, --star_fwhm STAR_FWHM
FWHM cut for the field stars. Given as i: FWHM_star < i x FWHM_target. Default 1.3
-cf CHECK_FILES, --check_files CHECK_FILES
Investigate files manually. "all":Check all files, "some": Only if an issue, "none": check none.
-ps [PLOT_STAMPS [PLOT_STAMPS ...]], --plot_stamps [PLOT_STAMPS [PLOT_STAMPS ...]]
Plot stamps of the apertures for each cycle
-bkg BKG_TYPE, --bkg_type BKG_TYPE
Either 2dmap or annulus
-ran R_ANNU [R_ANNU ...], --r_annu R_ANNU [R_ANNU ...]
Annulus BKG inner and outer radius in px (i.e. 5 10)
-sbkg [SAVE_BKG [SAVE_BKG ...]], --save_bkg [SAVE_BKG [SAVE_BKG ...]]
Save background 2dmap